# Apache Sling Interview Questions and Answers

## Basic Apache Sling Questions

**Q: What is Apache Sling?**

A: Apache Sling is a web framework that uses a Java Content Repository (JCR) to store and retrieve content. It maps HTTP request URLs to content resources using a resource-oriented architecture.

**Q: Why is Apache Sling used in AEM?**

A: Apache Sling provides the RESTful web framework and content resolution mechanism that Adobe Experience Manager (AEM) uses to deliver content. It enables dynamic content rendering and component binding.

**Q: How is Apache Sling different from traditional MVC frameworks?**

A: Traditional MVC frameworks map URLs to controllers, whereas Sling maps URLs to content resources and uses scripts or servlets to render them.

**Q: What is a Resource in Apache Sling?**

A: A Resource is a representation of a content node in the JCR. It is the central concept in Sling's resource-oriented architecture.

**Q: What is a ResourceResolver?**

A: ResourceResolver is an interface used to resolve paths to resources in the JCR. It also provides methods to adapt resources and access user session information.

**Q: What is the difference between Resource and Node?**

A: A Node is a JCR-level object representing a content item, while a Resource is a Sling abstraction over a Node, providing additional capabilities like adaptation.

**Q: What is Sling Adaptation?**

A: Sling Adaptation is a mechanism that allows converting one object type into another using the adaptTo() method.

**Q: What does adaptTo() do?**

A: The adaptTo() method attempts to convert an object (like a Resource or Request) into another type, such as a Sling Model or ValueMap.

**Q: What is a ValueMap?**

A: ValueMap is a map-like structure that provides easy access to properties of a resource.

**Q: What is a Sling Request?**

A: A Sling Request is an HttpServletRequest that includes additional Sling-specific methods and context, such as access to the Resource and ResourceResolver.

## Sling Request Processing

**Q: Explain the Sling request processing flow.**

A: Sling maps the request URL to a resource using the ResourceResolver. Then it determines the script or servlet to handle the request based on the resource type, selectors, and extension.

**Q: How does Sling resolve a request URL?**

A: Sling uses the ResourceResolver to map the URL path to a resource in the JCR. It then uses the resource type and selectors to find the appropriate script or servlet.

**Q: What are selectors in Sling?**

A: Selectors are optional parts of the URL that appear between the resource path and the extension. They allow different views or processing logic for the same resource.

**Q: What are extensions in Sling?**

A: Extensions indicate the desired response format, such as .html or .json.

**Q: What is a suffix in Sling URLs?**

A: A suffix is the part of the URL after the extension, often used to pass additional path information.

**Q: How are selectors, extensions, and suffix used together?**

A: They are used to determine the script or servlet to execute and to pass additional parameters or format hints.

**Q: What is the importance of HTTP methods in Sling?**

A: Sling uses HTTP methods (GET, POST, etc.) to determine the type of operation and to select the appropriate servlet.

**Q: How does Sling handle GET vs POST requests?**

A: GET is typically used for retrieving content, while POST is used for creating or modifying content. Sling provides different servlet interfaces for safe (GET) and all methods.

## Servlets & Scripts

**Q: What is a Sling Servlet?**

A: A Sling Servlet is a Java class that handles HTTP requests in Sling. It can be registered to handle specific paths or resource types.

**Q: How do you register a Sling Servlet?**

A: You can register a Sling Servlet using OSGi annotations like @SlingServletPaths or @SlingServletResourceTypes.

**Q: What is the difference between path-based and resourceType-based servlets?**

A: Path-based servlets are bound to specific URL paths, while resourceType-based servlets are bound to resources of a specific type.

**Q: What is a Sling Safe Methods Servlet?**

A: It is a servlet that handles safe HTTP methods like GET and HEAD.

**Q: What is a Sling All Methods Servlet?**

A: It is a servlet that handles all HTTP methods including POST, PUT, DELETE, etc.

**Q: How does Sling decide which servlet to execute?**

A: Sling uses the resource type, selectors, and HTTP method to find the best matching servlet.

**Q: What is servlet ranking in Sling?**

A: Servlet ranking determines the priority of servlets when multiple candidates match a request.

## Resource Types & Components

**Q: What is sling:resourceType?**

A: It is a property that defines the type of a resource and is used to resolve the appropriate script or servlet.

**Q: What is sling:resourceSuperType?**

A: It allows a resource type to inherit behavior from another resource type.

**Q: How does component inheritance work in Sling?**

A: Components can inherit scripts and logic from their super types using sling:resourceSuperType.

**Q: What is the script resolution order in Sling?**

A: Sling looks for scripts in the order of resource type, super type, and default handlers.

**Q: What happens if no matching servlet or script is found?**

A: Sling returns a 404 error or uses a default error handler.

## Sling Models

**Q: What are Sling Models?**

A: Sling Models are Java classes annotated with @Model that adapt from resources or requests and expose business logic.

**Q: How do Sling Models work internally?**

A: They use reflection and dependency injection to populate fields from the resource or request.

**Q: What are adaptables in Sling Models?**

A: Adaptables are the source objects (e.g., Resource, SlingHttpServletRequest) from which the model is adapted.

**Q: What is the difference between @Inject and @ValueMapValue?**

A: @Inject is a generic injection annotation, while @ValueMapValue specifically injects values from the resource's ValueMap.

**Q: What is @Self annotation?**

A: It injects the adaptable object itself into the model.

**Q: What is @Via annotation?**

A: It specifies an alternate path or method to resolve the injected value.

**Q: How do you make a Sling Model optional?**

A: Use the 'optional=true' attribute in the @Inject annotation.

**Q: How do you handle null values in Sling Models?**

A: Use Optional types or null checks in the model logic.

## Advanced Apache Sling

**Q: What is a Resource Provider?**

A: A Resource Provider is a service that provides access to resources from non-JCR sources.

**Q: What is a Sling Mapping?**

A: Sling Mapping is used to define URL rewriting rules for cleaner URLs.

**Q: What is /etc/map used for?**

A: It stores mapping configurations for internal to external URL translations.

**Q: What is Sling Rewriter?**

A: It is a framework for transforming HTML output using configurable pipelines.

**Q: What is Context-Aware Configuration in Sling?**

A: It allows configuration values to vary based on the context of the resource.

**Q: What is a Sling Filter?**

A: A Sling Filter is an OSGi service that intercepts HTTP requests and responses.

**Q: What are the different Sling Filter scopes?**

A: REQUEST, INCLUDE, FORWARD, ERROR.

**Q: How do you restrict access to Sling Servlets?**

A: Use OSGi annotations or Apache Sling's authentication and authorization mechanisms.

**Q: How does Sling support RESTful architecture?**

A: By mapping HTTP methods to operations and using resource-oriented URLs.

**Q: How does Sling handle content rendering?**

A: Using scripts and servlets mapped to resource types and request parameters.

## Scenario-Based / Practical Questions

**Q: How would you create a REST API using Apache Sling?**

A: Create a Sling Servlet registered with a resource type or path, and implement doGet/doPost methods to handle API logic.

**Q: How do you return JSON from a Sling Servlet?**

A: Set the response content type to application/json and write the JSON string to the response writer.

**Q: How do you handle exceptions in Sling Servlets?**

A: Use try-catch blocks and set appropriate HTTP status codes and error messages.

**Q: How do you improve Sling performance?**

A: Use caching, efficient queries, minimize resource resolution overhead, and optimize scripts.

**Q: How do you debug Sling request resolution issues?**

A: Use Sling logs, request.log, and the Sling Resource Resolver console (/system/console/resourceResolver).

**Q: How do you secure Sling-based endpoints?**

A: Use OSGi configurations, Apache Sling Referrer Filter, and authentication handlers.

**Q: How do you handle large content structures efficiently in Sling?**

A: Use pagination, lazy loading, and avoid deep hierarchies.